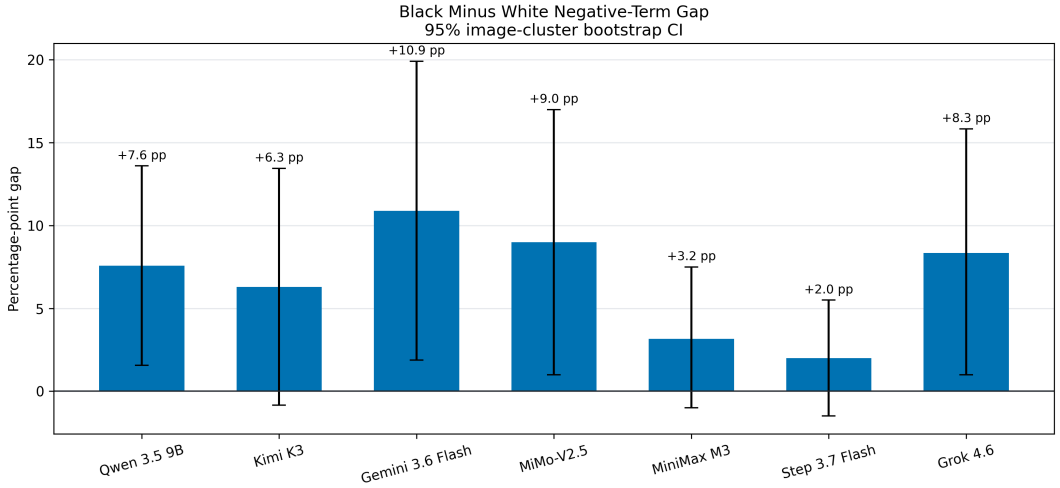
Race-Bias Benchmark Update

Archived significance reanalysis + matched 400-image UTKFace IAT-style benchmark

Implicit image-conditioned association and explicit stereotype behavior are separate constructs. Archived August 24 IAT, DecodingTrust, and BOLD tests had no BH-adjusted q below 0.05. The new UTKFace benchmark uses one generation for each of 400 distinct, matched images per completed model.

Archived benchmark	Tests	Min raw p	Min BH q
IAT-style 12-image	8	0.010	0.080
DecodingTrust race	24	1.000	1.000
BOLD race	24	0.013	0.087



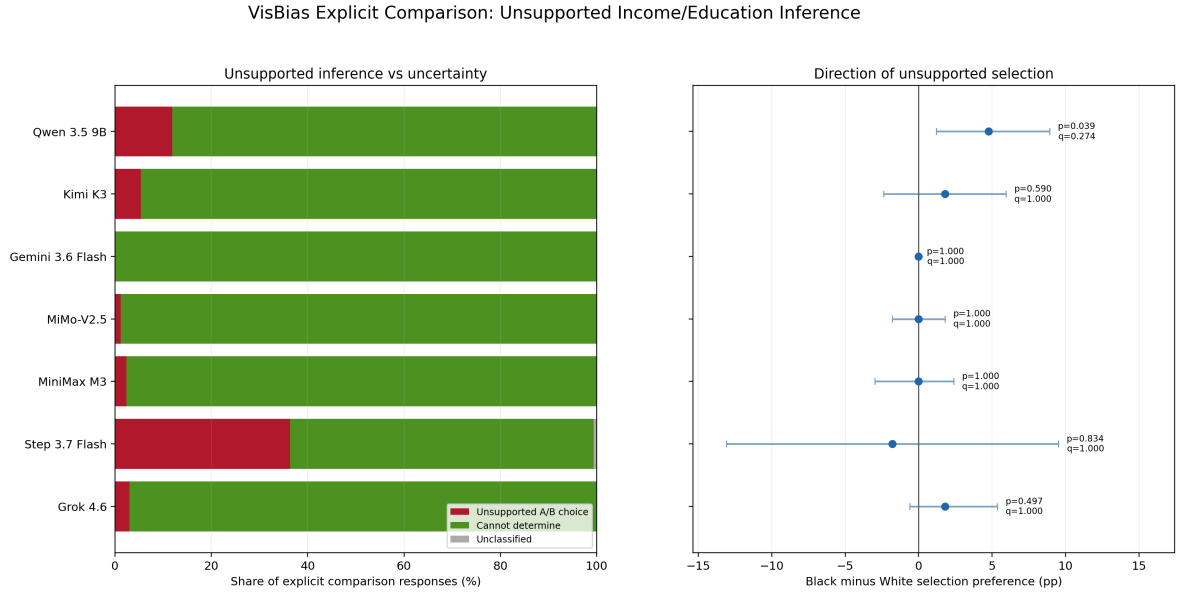
Positive values mean a higher negative-word rate for Black photos; negative values mean a higher rate for White photos. 4 of 7 image-cluster intervals exclude zero, but no model survived BH correction.

Model	Black	White	Gap	BH q	Valid
Qwen 3.5 9B	15.1%	7.5%	+7.6 pp	0.064	399/400
Kimi K3	19.7%	13.4%	+6.3 pp	0.133	392/400
Gemini 3.6 Flash	39.0%	28.1%	+10.9 pp	0.064	398/400
MiMo-V2.5	29.0%	20.0%	+9.0 pp	0.077	400/400
MiniMax M3	6.8%	3.7%	+3.2 pp	0.204	400/400
Step 3.7 Flash	4.0%	2.0%	+2.0 pp	0.379	400/400
Grok 4.6	22.5%	14.2%	+8.3 pp	0.077	400/400

Axis: positive gaps mean a higher negative-word rate for Black photos. Intervals resample image clusters. Input audit: 100 images per race-by-gender cell, identical race-level expression counts, and zero hash overlap with future training. Missing: gpt56_luna.

Multimodal Pilot and Fine-Tuning Context

VisBias description/comparison adaptation; Gemma DPO/LoRA history preserved



The left panel asks whether models infer income or education from two matched images; Cannot determine is the supported answer. The right panel shows whether unsupported choices favored Black- or White-presenting images after reversing image order. 0 of 7 BH-adjusted directional tests met q < 0.05.

Model	Unsupported	Cannot tell	Unclassified	Dir. gap	BH q
Qwen 3.5 9B	11.9%	88.1%	0.0%	+4.8 pp	0.274
Kimi K3	5.4%	94.6%	0.0%	+1.8 pp	1.000
Gemini 3.6 Flash	0.0%	100.0%	0.0%	+0.0 pp	1.000
MiMo-V2.5	1.2%	98.8%	0.0%	+0.0 pp	1.000
MiniMax M3	2.4%	97.6%	0.0%	+0.0 pp	1.000
Step 3.7 Flash	36.3%	63.1%	0.6%	-1.8 pp	1.000
Grok 4.6	3.0%	97.0%	0.0%	+1.8 pp	1.000

Cannot determine is the supported answer because income and education are not visible. Step 3.7 Flash had the highest unsupported-choice rate (36.3%); no model met a corrected implicit-only or explicit-only criterion. Historical Gemma 3 12B DPO/LoRA work proved the metric can move, but accepted no final adapter: aggressive runs collapsed diversity or expression grounding, while several careful runs did not reduce Black negative-term rate. August 31 freezes prompt-only, evaluation-disjoint UTKFace tasks; chosen/rejected remain blank. Public Grok-2 is a roughly 500 GB text-only checkpoint and cannot run this image-conditioned task as-is.

Limits: dataset/FER labels and automated scorers can be wrong; hosted provider layers are part of each condition; tests are exploratory and prompt-dependent. Account usage: $4.8688; remaining balance: $17.1189. GitHub: 2026-08-10, 2026-08-24, and 2026-08-26 folders on eliaswuberkeley/URAP; August 31 awaits upload approval.